HOLOMORPHIC TWISTS AND AN ORDINARY FRIED OBSTRUCTION NEAR HYPERBOLIC METRICS

CHATGPT

Abstract. We construct an obstruction to the ordinary Fried identity arbitrarily close to a hyperbolic three-metric. Along negatively curved metrics converging to it and acyclic unitary characters tending to the trivial character, the Ruelle zeta value divided by analytic torsion squared tends to zero. A resonance crossing at quadratic character scale drives the limit, controlled by uniform character holomorphy and finite-cut torsion variation.

AI statement. Simon Becker reports that the original argument was obtained from a single prompt containing only the conjecture, submitted to GPT-6.1 Sol with Ultra reasoning in Codex. Subsequent prompts revised the exposition and references.

1. Introduction

Let (Σ, g) be a closed oriented negatively curved Riemannian manifold of dimension $n > 2$, and let χ be an acyclic unitary representation of $\pi_1(\Sigma)$. Write Id for the identity endomorphism and $\mathcal{G}_g^{\#}$ for the primitive oriented periodic orbits of the geodesic flow; both orientations of each closed geodesic are included. For $\gamma \in \mathcal{G}_g^{\#}$, let $\ell_g(\gamma)$ be its length and $\chi(\gamma)$ its holonomy. For a complex variable s, the Ruelle zeta function in the convention used here is

$$Z_{g,\chi}(s) = \prod_{\gamma \in \mathcal{G}_g^{\#}} \det\bigl(\mathrm{Id} - \chi(\gamma) e^{-s\ell_g(\gamma)}\bigr), \qquad \operatorname{Re} s \gg 1. \tag{1}$$

Let $T_\chi(\Sigma)$ denote the Ray–Singer analytic torsion of the local system defined by χ. Conjecture 9 in Zworski's survey [1] asserts

$$|Z_{g,\chi}(0)|^{(-1)^{n-1}} = T_\chi(\Sigma)^2. \tag{2}$$

In dimension three, the exponent in (2) is $+1$. The spectral coordinate used in the survey is a complex variable λ; its zeta function, denoted by ζ_χ, is related to ours by

$$\zeta_\chi(\lambda) = Z_{g,\chi}(-i\lambda).$$

The value at zero requires separate attention when a resonance reaches the origin. A zero resonance can contribute both numerator and denominator factors to the dynamical determinant. The question is therefore the value of the specialized meromorphic function, after its spectral factors have been cancelled.

We write $\mathbf{1}$ for the trivial character. The conclusion of the argument is the following.

Theorem 1.1. *There exist a closed oriented hyperbolic three-manifold (Σ, g_0), negatively curved metrics $g_j \to g_0$ in C^∞, and acyclic unitary line characters $\chi_j \to \mathbf{1}$ such that Z_{g_j,χ_j} is holomorphic at zero and*

$$0 \le \frac{|Z_{g_j,\chi_j}(0)|}{T_{\chi_j}(\Sigma)^2} \longrightarrow 0. \tag{3}$$

Consequently the asserted identity (2) fails for every sufficiently large j.

Date: 11 October 2026; research proof draft.

Key words and phrases. Fried conjecture, Ruelle zeta function, dynamical torsion, contact Anosov flow, holomorphic character deformation.

The decisive steps are uniform holomorphy on a fixed character disk and the extension of the finite-cut variation calculation to a cut containing nonzero resonances. The latter extension is proved explicitly in Section 3.

The historical results fix both the benchmark and the scope of the question. Ray and Singer [2] defined analytic torsion from the twisted Hodge spectrum and conjectured its equality with Reidemeister–Franz torsion. Let $\Delta_{\chi,k}$ be the positive twisted Hodge Laplacian on degree-k forms. Write $\zeta_{\Delta_{\chi,k}}$ for its spectral zeta function, formed from the positive eigenvalues and continued to zero, and define $\det{}'(\Delta_{\chi,k}) = \exp(-\zeta'_{\Delta_{\chi,k}}(0))$; the prime excludes zero eigenvalues. Our convention is

$$\log T_\chi(\Sigma) = \frac{1}{2}\sum_{k=0}^{\dim\Sigma}(-1)^k k\,\zeta'_{\Delta_{\chi,k}}(0), \qquad T_\chi(\Sigma) = \prod_k \det{}'(\Delta_{\chi,k})^{(-1)^{k+1}k/2}. \tag{4}$$

For an acyclic unitary system this is independent of the Riemannian metric. Cheeger [3] and Müller [4] established the equality with combinatorial torsion; Müller later extended the comparison to unimodular representations [5]. These results make torsion a fixed benchmark along the metric deformation. They do not identify a dynamical zeta value.

Fried's hyperbolic theorem [6] supplies that identification for closed oriented hyperbolic manifolds with acyclic orthogonal coefficients. For an odd-dimensional base, the ordinary oriented-orbit product is regular and nonzero at the origin, with modulus equal to the square of the base torsion. His nonacyclic theorem instead determines a leading Laurent term. A complex unitary local system can be treated by realification: its real torsion is the square of its complex torsion, and its real zeta function is the product of the complex zeta functions for the representation and its conjugate. Reversal of oriented geodesics identifies these latter functions. Taking the positive square root gives the complex-unitary normalization used here. This translation supplies the hyperbolic calibration, without any uniform statement near the trivial character.

The square also reflects the distinction between the base Σ and the flow manifold $M = S\Sigma$, its unit tangent sphere bundle. Let $\pi : M \to \Sigma$ be the bundle projection, and use $H^\bullet(\Sigma;\chi)$ for cohomology with coefficients in the local system defined by χ. For an oriented base of dimension at least three, the sphere-bundle homotopy sequence identifies the fundamental groups of Σ and $S\Sigma$. Pullback preserves acyclicity, and in odd base dimension the torsion comparison is

$$H^\bullet(\Sigma;\chi) = 0 \quad\Longleftrightarrow\quad H^\bullet(S\Sigma;\pi^*\chi) = 0, \qquad T_{\pi^*\chi}(S\Sigma) = T_\chi(\Sigma)^2. \tag{5}$$

See [7, Propositions 5.1–5.2]. General flow formulations therefore compare with torsion on $S\Sigma$, whereas (2) is stated on the base. Here these manifolds have dimensions five and three, respectively.

Fried's later formulations have related but different hypotheses. *Lefschetz formulas for flows* [8] develops the locally homogeneous setting with orbit indices; the conjecture on page 66 is stated in that setting. His analytic-flow paper [9] proposes, on page 181, a torsion identity for negatively curved real analytic geodesic flows using matrix coefficients on the unit tangent bundle and orientation factors. Translating coefficient assumptions, orbit weights, and torsion conventions is necessary before identifying these statements. The precise smooth, unitary, ordinary-value assertion addressed by Theorem 1.1 is Zworski's Conjecture 9.

The locally symmetric literature gives a larger positive class. Moscovici–Stanton [10] developed a trace-formula proof. Let G be a connected linear real reductive group, K a maximal compact subgroup, and $\Gamma \subset G$ a torsion-free cocompact discrete subgroup. Shen [11] established the unitary theorem for closed odd-dimensional locally symmetric spaces $\Gamma\backslash G/K$. His leading Laurent expansion has order zero and torsion-squared coefficient when the bundle is acyclic. The proof uses semisimple orbital integrals and representation theory, and completes an argument in the earlier work. It covers the hyperbolic starting metric, while the conformal family below leaves the locally symmetric class. Shen's zeta and torsion conventions are both reciprocal to those used here [11, (1-4)–(1-5)]; both must be translated. Locally symmetric formulas also retain Fuller-index weights when closed geodesics occur in families.

Bunke–Olbrich [12, 13] connect Selberg functions to elliptic operators on compact rank-one locally symmetric spaces. In the odd-dimensional hyperbolic case, their determinant formulas express Selberg factors through regularized elliptic determinants and explicit nonvanishing exponential factors. The middle-degree Weyl pair requires a graded determinant. These formulas identify local zeros and their multiplicities in the appendix; Fried's theorem separately fixes the modulus on the hyperbolic axis. In particular, local determinant enumeration and global torsion normalization play different roles.

The continuation theory explains why the remaining issue is a special value. Rugh [14] used generalized Fredholm determinants for analytic hyperbolic dynamics in dimension three, and Fried [9] treated analytic hyperbolic flows in arbitrary dimension. Giulietti–Liverani–Pollicott [15] obtained meromorphic continuation to the whole plane for smooth Anosov flows using transfer operators on anisotropic spaces of currents. Their erratum [16] corrects a contact spectral-gap argument without affecting meromorphic continuation. None of these continuation results alone ensures regularity, nonvanishing, or a torsion identity at zero.

The microlocal construction is especially suited to the present deformation. Faure–Sjöstrand [17] built anisotropic Sobolev spaces with discrete generator spectrum in suitable half-planes; Dyatlov–Zworski [18] combined Fredholm theory, propagation, and the flat trace to obtain a microlocal continuation proof. Generalized resonant states can be distributions, and the ordinary divisor is an alternating combination of their multiplicities. Guedes Bonthonneau [19] constructed common anisotropic spaces for nearby Anosov fields, giving continuous finite clusters and smooth individual resonances away from collisions. At a collision the natural objects are the contour projector and characteristic polynomial. Cluster continuity alone neither selects smooth roots nor controls a coefficient relative to a denominator tending to zero.

Dang–Rivière [20, Theorem 2.1] give the relevant topological interpretation. For a flat vector bundle E with connection ∇ over M, let d_∇ be its flat de Rham differential and $C^\bullet(0)$ the full generalized zero-resonance complex. Projection onto that complex induces

$$H^k(C^\bullet(0), d_\nabla) \simeq H^k(M; E). \tag{6}$$

This is a statement about the full de Rham complex, including generalized states. It does not identify each horizontal resonance space with a cohomology group. In particular,

$$H^\bullet(M; E) = 0 \quad \Longrightarrow \quad C^\bullet(0) \text{ is exact}, \qquad C^\bullet(0) \text{ exact } \not\Longrightarrow \ C^\bullet(0) = 0.$$

A nonzero exact complex can contribute torsion when a resonance reaches zero. This distinction is used in the crossing argument rather than assuming horizontal exactness or semisimplicity.

Low-dimensional results give useful precedents with different ambient dimensions. Write $\chi_{\mathrm{top}}(\Sigma)$ for the Euler characteristic. Dyatlov–Zworski [21] proved that the untwisted surface-geodesic zeta function has order $-\chi_{\mathrm{top}}(\Sigma)$ at zero, and also treated contact Anosov flows on three-dimensional flow manifolds. Cekić–Paternain [22] studied volume-preserving flows in that dimension, including unitary twists, twisted semisimplicity, and an untwisted time-change example with a zero Jordan block. Their later work [23] treats dissipative flows and Gaussian thermostats with changed untwisted zeta order. An arbitrary time change or thermostat need not be a Riemannian geodesic flow. These examples illuminate the role of generalized multiplicities, but do not themselves supply an acyclic unitary obstruction on the five-dimensional flow manifold here.

Dang–Guillarmou–Rivière–Shen [24] prove local constancy of the ordinary nonzero value where zero is absent from the full resonant spectrum. Their Theorem 3 specializes this to geodesic flow over a hyperbolic three-manifold: for a fixed acyclic unitary representation, Selberg factorization excludes zero, and Fried's identity persists for nearby Anosov fields. The representation is fixed before the neighborhood is chosen. Denote the resulting metric neighborhood of g_0 by $\mathcal{V}_\chi$. In schematic quantifier form, the conclusion is

$$\text{for each fixed acyclic unitary } \chi, \quad \exists\, \mathcal{V}_\chi \ni g_0 : \quad |Z_{g,\chi}(0)| = T_\chi(\Sigma)^2 \quad (g \in \mathcal{V}_\chi). \tag{7}$$

It does not assert one neighborhood valid for all characters tending to the nonacyclic trivial character. The simultaneous limit in Theorem 1.1 therefore lies outside that uniform conclusion.

Cekić–Delarue–Dyatlov–Paternain (CDDP) [25] establish the untwisted instability that supplies the geometric input. On a hyperbolic three-manifold with first Betti number $b_1 = \dim H^1(\Sigma;\mathbb{R})$, the horizontal generalized zero ranks are $(1, 2b_1, 2b_1+2, 2b_1, 1)$. Under their nondegeneracy condition, small nonzero perturbations have semisimple zero ranks $(1, b_1, b_1+2, b_1, 1)$; their conformal argument makes this condition generic. Accordingly the ordinary vanishing order changes from $4-2b_1$ to $4-b_1$. For $b_1 = 1$, the initial ranks are precisely $(1, 2, 4, 2, 1)$. These are nonacyclic untwisted statements. The present construction must additionally follow the twisted cluster and control its specialized value at a character-dependent crossing.

Chaubet–Dang [26] account for zero resonances by defining dynamical torsion, denoted by τ_{dyn}. Equation numbers for this reference follow the linked author version v2. Write Z for the chosen Ruelle function, m for its order at zero, and $a \neq 0$ for its leading Laurent coefficient, so that $Z(s) = a\, s^m(1 + O(s))$. For a contact form α on a manifold of odd dimension $2q+1$, so that $q = (\dim M - 1)/2$, let Γ_α denote the induced chirality, a degree-reversing involution; its explicit formula is given in Section 3. In the acyclic case, write $\tau(C^\bullet(0), \Gamma_\alpha)$ for the corresponding refined torsion of the finite zero complex, and put

$$Q_0 = \sum_{k=0}^{q} (-1)^k (q+1-k) \dim C^k(0).$$

Their zero-cut comparison reads

$$\tau_{\mathrm{dyn}} = (-1)^{Q_0} a^{(-1)^q} \tau\big(C^\bullet(0), \Gamma_\alpha\big). \tag{8}$$

See [26, (6.7)]; here $q = 2$, hence the exponent of a is $+1$. Their construction is independent of the cut, invariant under contact Anosov deformations, and holomorphic on the acyclic representation locus; its quotient by suitable Turaev torsion is locally constant. The ordinary value $Z(0)$, the leading coefficient a, and the corrected torsion are distinct objects. Even when $m = 0$, a nontrivial zero complex can prevent replacing (8) by $\tau_{\mathrm{dyn}} = Z(0)$.

Bismut–Shen [27] extend this perspective to a canonical nonzero Fried section of the cohomological determinant line, denoted by $\det H^\bullet(M; E)$, for general Anosov fields, without excluding zero resonances. Their section is C^1 in families and agrees with the contact construction. Gauss–Manin flatness requires the bundle to be flat on the total fibration; a family of fiberwise flat connections alone does not give that hypothesis. Their norm-one identity for unimodular coefficients remains a modified conjecture. Thus extending a regularized torsion through a finite spectral cut already has substantial precedents. The scalar ordinary-value calculation at the exceptional crossing is the issue here.

Recent generic results sharpen this distinction. In its March 2026 revision, Humbert–Tao [28] prove a fixed-representation dichotomy on a connected open set of Anosov metrics: semisimplicity and closedness of all horizontal zero states either hold on an open dense set or hold nowhere. In the first case the dimensions are determined by twisted cohomology. Their Corollary 1.2 extends the untwisted generic description to the Anosov component containing a hyperbolic three-metric. Their surface results also treat generic nonunitary representations and produce exceptional acyclic twists with zero Jordan blocks. Generic conclusions for fixed twists permit exceptional metric loci; they do not give uniform control as a character approaches the trivial representation, or assert the ordinary value at every exceptional metric.

Other coefficient and flow classes require additional translations. For nonunitary twists on odd-dimensional hyperbolic manifolds, Spilioti [29] compares the Ruelle value with complex-valued torsions for acyclic representations sufficiently close to the acyclic unitary locus. Shen [30] develops the locally symmetric comparison with Cappell–Miller torsion and, more generally, leading terms. These conclusions concern complex torsion and locally symmetric metrics; their neighborhoods do

not include a uniform passage to a nonacyclic character. Jézéquel–Zung [31] prove a Fried identity for transitive smooth pseudo-Anosov flows on closed three-dimensional flow manifolds, with acyclic unitary coefficients and explicit monodromy conditions at singular orbits (or a monodromy condition when there are no singular orbits). Shen–Yu [32] relate Morse–Smale dynamics, Milnor metrics, and torsion, with connecting trajectories contributing to the comparison. These results broaden the dynamical setting but do not cover the five-dimensional negatively curved geodesic flow considered here.

The proposed obstruction consequently concerns a precise remaining distinction: invariance of corrected dynamical torsion does not by itself identify the ordinary specialized zeta value at a nonzero exact zero complex. The argument below uses the published hyperbolic calibration, perturbation framework, cohomological theorem, and untwisted crossing mechanism. Its additional work is the uniform complex character disk and the finite-cut normalization estimate at quadratic character scale. Those estimates, rather than continuation or regularization alone, are what must establish Theorem 1.1.

For the construction, let $b_1(\Sigma) = \dim H^1(\Sigma;\mathbb{R})$ be the first Betti number and choose the hyperbolic base so that $b_1(\Sigma) = 1$. Let ω be a nonzero real harmonic one-form, d the exterior differential, and b a smooth real conformal direction on Σ. For a small real metric parameter $\tau \geq 0$ and a complex character parameter z, put

$$g_\tau = e^{-2\tau b} g_0, \qquad d_z = d + iz\omega\wedge, \qquad \chi_z(\gamma) = \exp\left(-iz\int_\gamma \omega\right), \qquad Z(s,z,\tau) = Z_{g_\tau,\chi_z}(s).$$

We use θ for real values of z, so that χ_θ is unitary. Denote the two small horizontal degree-one roots of the negative twisted Lie derivative by $c(z,\tau)$ and $t(z,\tau)$. Set $\kappa = \|\omega\|^2_{L^2(\Sigma,g_0)}/\operatorname{vol}_{g_0}(\Sigma) > 0$, and denote by $r > 0$ the initial metric derivative of the companion root $t(0,\tau)$ for the chosen direction. At the hyperbolic metric the two roots coincide, while the untwisted metric variation moves the companion root:

$$\begin{aligned} c(\theta,0) = t(\theta,0) &= -\tfrac{1}{2}\kappa\theta^2 + O(\theta^4), \qquad \kappa > 0, \\ t(0,\tau) &= r\tau + o(\tau), \qquad r > 0. \end{aligned} \tag{9}$$

Write $\sigma_\theta > 0$ for a crossing parameter, characterized by $t(\theta,\sigma_\theta) = 0$; it satisfies $\sigma_\theta \sim \kappa\theta^2/(2r)$. After the degree-one factors are removed, the degree-two characteristic polynomial has a residual quadratic factor $Q(s,z,\tau)$. Denote its linear coefficient by $-e_1(z,\tau)$ and its constant coefficient by $e_2(z,\tau)$, so that $Q(s,z,\tau) = s^2 - e_1(z,\tau)s + e_2(z,\tau)$. Let $G(s,z,\tau)$ be the holomorphic nonvanishing factor remaining after extraction of the finite resonance divisor, constructed in Section 3. At a crossing, exactness of the full zero complex forces $e_2 = 0$. With c and e_1 evaluated at (θ,σ_θ), the ordinary divisor becomes

$$Z(s,\theta,\sigma_\theta) = G(s,\theta,\sigma_\theta)(s-c)(s-e_1).$$

The three estimates responsible for (3) are

$$\frac{G(0,\theta,\sigma_\theta)}{G(0,\theta,0)} \longrightarrow 1, \qquad \frac{c(\theta,\sigma_\theta)}{c(\theta,0)} \longrightarrow 1, \qquad \frac{e_1(\theta,\sigma_\theta)}{c(\theta,0)} \longrightarrow 0. \tag{10}$$

Section 2 constructs the common cut, proves the coefficient estimates, and produces the crossing. Section 3 controls the fixed-cut normalization and specializes the ordinary zeta function to prove Theorem 1.1. The appendix records the geometric and spectral inputs in their source conventions.

2. The spectral cut and the crossing

2.1. **Geometric setup and hyperbolic normalization.** Let (Σ, g_0) be a closed connected oriented hyperbolic three-manifold with $b_1 = 1$, and let $\omega \neq 0$ be its real harmonic one-form. Choose a

smooth real function b on Σ, to be specified below, and a real metric parameter $\tau \geq 0$ near zero. With d denoting the exterior differential, set

$$d_z = d + iz\omega\wedge, \qquad g_\tau = e^{-2\tau b} g_0, \qquad z \in \mathbb{C},$$

and identify the unit bundles by radial scaling with $M = S_{g_0}\Sigma$:

$$F_\tau(x, v) = (x, e^{\tau b(x)} v), \qquad F_\tau : M \longrightarrow S_{g_\tau}\Sigma.$$

Write $\pi : M \to \Sigma$ for the bundle projection. The character of the flat connection on the trivial complex line bundle is $\chi_z(\gamma) = \exp(-iz\int_\gamma \omega)$. On M the same notation d_z denotes the pullback differential $d + iz\pi^*\omega\wedge$. Write α_τ, X_τ for the pulled-back geodesic contact form and Reeb field. If ι_X denotes contraction with a vector field X, the twisted Lie derivative and its negative are

$$L_{z,\tau} = \mathcal{L}^{d_z}_{X_\tau} = d_z\iota_{X_\tau} + \iota_{X_\tau} d_z, \qquad B_{z,\tau} = -L_{z,\tau}.$$

We suppress (z, τ) on operators when the parameter is clear. A form is called horizontal when $\iota_{X_\tau} u = 0$. The spectral coordinate s is an eigenvalue of B:

$$L + s = s\,\mathrm{Id} - B, \qquad (L + s)^{-1} = (s\,\mathrm{Id} - B)^{-1}.$$

For comparison with CDDP's convention, $P = -iL = iB$ and $B = -iP$. For real $z = \theta \neq 0$ sufficiently small, the unitary character is acyclic. The construction of the base and the Hodge proof of acyclicity are given in the appendix.

The conformal direction is chosen so that the untwisted horizontal degree-one companion root has derivative $r > 0$ at $\tau = 0$. Let Δ_{g_0} be the nonpositive scalar Laplacian and Q_4 the self-adjoint periodized smoothing operator of CDDP's Theorem 5. One explicit choice is $b = Q_4\Delta_{g_0}|\omega|^2$. The appendix defines this operator in (80) and verifies the nonvanishing pairing and the positive sign using CDDP's contact-orientation convention. The actual primary input is CDDP (3.55)–(3.57), (4.22), (4.38), Theorem 5, and Proposition A.1 [25]. With $b_1 = 1$, this is the nondegeneracy condition in CDDP Theorem 3. Their compressed $P = -iL$ variation is multiplied by $-i$ to obtain the variation of $B = -iP$.

Fix $\lambda > 0$ sufficiently small so that the positively oriented circle $\gamma = \{|s| = \lambda\}$ contains no boundary resonances at $(z, \tau) = (0, 0)$ in any degree. The horizontal cut $\mathcal{U}^k_{z,\tau}$ is the sum of the generalized resonance spaces of $B_{z,\tau}$ in degree k inside γ, restricted to horizontal forms. Its ranks in degrees $0, \ldots, 4$ are

$$(1, 2, 4, 2, 1), \qquad \text{and its full ranks are } (1, 3, 6, 6, 3, 1). \tag{11}$$

The full ranks refer to degrees $0, \ldots, 5$, with no horizontality restriction. Define the compressed operators, scalar root, companion root, and characteristic polynomials by

$$A_k(z, \tau) = B_{z,\tau}|_{\mathcal{U}^k_{z,\tau}}, \qquad c(z, \tau) = A_0(z, \tau), \qquad t(z, \tau) = \operatorname{tr} A_1(z, \tau) - c(z, \tau),$$

$$P_k(s, z, \tau) = \det(s\,\mathrm{Id} - A_k(z, \tau)).$$

The rank-one operator A_0 is identified with its scalar root c. Let $\mu_0(\theta)$ be the small positive twisted scalar Hodge eigenvalue, and put $c_0(\theta) = -1 + \sqrt{1 - \mu_0(\theta)}$, using the square-root branch equal to one at the origin. The Selberg enumeration in the appendix, including its scalar denominator cancellation, gives on the hyperbolic axis, for small real $\theta \neq 0$,

$$c(\theta, 0) = t(\theta, 0) = c_0(\theta) = -1 + \sqrt{1 - \mu_0(\theta)} < 0, \qquad \det(s - A_2(\theta, 0)) = (s - c_0)^3(s + c_0). \tag{12}$$

These are the consequences of DGRS (7.6) and its Selberg normalization used here [24]; they are not deduced from abstract contact algebra.

For later reference,

$$\mu_0(\theta) = \kappa\theta^2 + O(\theta^4), \qquad c_0(\theta) = -\tfrac{1}{2}\kappa\theta^2 + O(\theta^4), \qquad \kappa = \frac{\|\omega\|^2_{L^2(\Sigma)}}{\operatorname{vol}(\Sigma)} > 0. \tag{13}$$

Let d^* denote the g_0-adjoint of d. The first derivative of the scalar Laplacian annihilates the normalized constant, since $d^*\omega = 0$. The first correction to that simple normalized eigenfunction is therefore zero. The coefficient of θ^2 in its eigenvalue is the expectation of $|\omega|^2$. Conjugation under $\theta \mapsto -\theta$ makes this eigenvalue even. Expanding the square root gives the second assertion.

2.2. **Uniform holomorphy and reversal symmetry.**

Lemma 2.1. *Write* $\mathrm{Hol}(D)$ *for the holomorphic functions on a complex domain* D*. After shrinking the parameter neighborhood, there are* $\varepsilon > 0$ *and one fixed* $\rho > 0$ *such that the finite matrices for the cut and their characteristic coefficients belong to*

$$C^0\big([0,\varepsilon]_\tau; \mathrm{Hol}(\{|z| < \rho\})\big).$$

The topology is locally uniform convergence in the complex z *variable. Their ranks are* (11)*.*

Proof. In each form degree, fix a finite collection of anisotropic spaces $\mathcal{H}_0, \dots, \mathcal{H}_4$ for the untwisted metric family, as in [26, Appendix B.4–B.6]. Write $R_{z,\tau}(s) = (L_{z,\tau} + s)^{-1}$ for the resolvent, suppressing the form-degree index. For each degree and each of these levels, the untwisted contour resolvent satisfies

$$\sup_{\substack{s\in\gamma \\ 0\le\tau\le\varepsilon}} \|R_{0,\tau}(s)\|_{\mathcal{H}_j\to\mathcal{H}_j} < \infty, \qquad R_{0,\tau}(s) = (L_{0,\tau} + s)^{-1}. \tag{14}$$

The twist is an order-zero multiplication operator, as in [26, (8.9)]:

$$L_{z,\tau} = L_{0,\tau} + zV_\tau, \qquad V_\tau = i\,\pi^*\omega(X_\tau), \qquad \sup_\tau \|V_\tau\|_{\mathcal{H}_j\to\mathcal{H}_j} < \infty. \tag{15}$$

Choose one radius $\rho > 0$ so small that, simultaneously on this finite list of spaces,

$$\sup_{\substack{|z|<\rho,\ s\in\gamma \\ 0\le\tau\le\varepsilon}} \|zV_\tau R_{0,\tau}(s)\|_{\mathcal{H}_j\to\mathcal{H}_j} \le \tfrac{1}{2}. \tag{16}$$

The factorization

$$L_{z,\tau} + s = (\mathrm{Id} + zV_\tau R_{0,\tau}(s))(L_{0,\tau} + s)$$

then yields the convergent Neumann series

$$R_{z,\tau}(s) = R_{0,\tau}(s)(\mathrm{Id} + zV_\tau R_{0,\tau}(s))^{-1} = \sum_{n=0}^{\infty} (-z)^n R_{0,\tau}(s)(V_\tau R_{0,\tau}(s))^n. \tag{17}$$

This proves holomorphy in the same disk for every nearby metric and gives uniform bounds for the twisted resolvent.

Metric continuity is used with a loss of one level. Let C_j bound the contour resolvent norms on both levels j and $j+1$, uniformly in the chosen parameter neighborhood. Then

$$R_{z,\tau}(s) - R_{z,\tau'}(s) = R_{z,\tau}(s)(L_{z,\tau'} - L_{z,\tau})R_{z,\tau'}(s), \tag{18}$$
$$\|R_{z,\tau}(s) - R_{z,\tau'}(s)\|_{\mathcal{H}_{j+1}\to\mathcal{H}_j} \le C_j^2 \|L_{z,\tau'} - L_{z,\tau}\|_{\mathcal{H}_{j+1}\to\mathcal{H}_j} \longrightarrow 0.$$

The last convergence is uniform for $s \in \gamma$ and for z in any smaller closed disk. Define the full-cut spectral projection $\Pi_{z,\tau}$ and its first spectral moment $\mathsf{T}_{z,\tau}$ by contour integration:

$$\Pi_{z,\tau} = \frac{1}{2\pi i}\int_\gamma R_{z,\tau}(s)\,ds, \qquad \mathsf{T}_{z,\tau} = \frac{1}{2\pi i}\int_\gamma sR_{z,\tau}(s)\,ds = B_{z,\tau}\Pi_{z,\tau}. \tag{19}$$

They have the same holomorphic and continuous dependence.

The untwisted ranks persist by [26, Appendix B.6]. After shrinking ρ, the Neumann series also gives

$$\|\Pi_{z,\tau} - \Pi_{0,\tau}\|_{\mathcal{H}_j\to\mathcal{H}_j} < 1,$$

so the two finite-rank projections have equal ranks. To pass from operators to finite matrices, let N be the cut rank in the chosen form degree and choose fixed smooth sources $f_1, \dots, f_N$ and fixed smooth dual tests $\ell_1, \dots, \ell_N$, viewed as linear functionals on that degree, such that

$$F_{ij}(z,\tau) = \ell_i(\Pi_{z,\tau} f_j), \qquad K_{ij}(z,\tau) = \ell_i(\mathsf{T}_{z,\tau} f_j), \qquad \det F(0,0) \neq 0. \tag{20}$$

Then $\det F(z,\tau) \neq 0$ nearby, the vectors $\Pi_{z,\tau} f_j$ form a frame, and the compressed operator has matrix

$$[B_{z,\tau}]_{\{\Pi f_j\}} = F(z,\tau)^{-1} K(z,\tau). \tag{21}$$

This formula makes the asserted regularity of every characteristic coefficient explicit. For the horizontal degrees one first inserts $h_\tau = \mathrm{Id} - \alpha_\tau \wedge \iota_{X_\tau}$, which commutes with the resolvent and identifies the horizontal bundles. No same-level operator-norm continuity in τ is required. □

Lemma 2.2. *The characteristic coefficients in Lemma 2.1 are even in z and real for real z. In particular c is even and real on the real locus.*

Proof. The reversal argument applies to each horizontal degree separately. For a primitive oriented orbit γ, let ℓ_γ be its length, $\bar\gamma$ its reverse, P_γ its full transverse symplectic return map, and $I_\gamma = \int_{\pi\gamma} \omega$. Write Λ^k for the kth exterior power. In a right half-plane, the degree-k dynamical determinant $\mathscr{D}_k$ is

$$\mathscr{D}_k(s,z,\tau) = \exp\left[- \sum_{\gamma \text{ primitive}} \sum_{j \geq 1} \frac{e^{-sj\ell_\gamma}}{j} \frac{e^{-izjI_\gamma} \operatorname{tr}(\Lambda^k P_\gamma^j)}{|\det(\mathrm{Id} - P_\gamma^j)|} \right]. \tag{22}$$

Let φ_t be the geodesic flow of X_τ. The reversal $R(x,v) = (x,-v)$ satisfies, with $\sim$ denoting similarity of linear maps,

$$R\varphi_t = \varphi_{-t} R, \qquad I_{\bar\gamma} = -I_\gamma, \qquad P_{\bar\gamma} \sim P_\gamma^{-1}.$$

If $\mathbb{J}$ is the standard symplectic matrix in a symplectic basis, then

$$(P_\gamma^j)^T \mathbb{J} P_\gamma^j = \mathbb{J} \quad \Longrightarrow \quad P_\gamma^{-j} = \mathbb{J}^{-1} (P_\gamma^j)^T \mathbb{J}.$$

Consequently, for every k,

$$\operatorname{tr}(\Lambda^k P_\gamma^{-j}) = \operatorname{tr}(\Lambda^k P_\gamma^j), \qquad |\det(\mathrm{Id} - P_\gamma^{-j})| = |\det(\mathrm{Id} - P_\gamma^j)|. \tag{23}$$

Pairing the two orientations therefore replaces the holonomy factor by

$$e^{-izjI_\gamma} + e^{izjI_\gamma} = 2\cos(zjI_\gamma).$$

The paired Euler sum gives

$$\mathscr{D}_k(s,-z,\tau) = \mathscr{D}_k(s,z,\tau), \qquad \mathscr{D}_k(\bar s,\theta,\tau) = \overline{\mathscr{D}_k(s,\theta,\tau)} \quad (\theta \in \mathbb{R}). \tag{24}$$

These identities continue with the determinants. Their zero divisors are the degreewise resonance multisets, including algebraic multiplicity. The monic characteristic polynomial of the fixed, conjugation-invariant cut is uniquely determined by that multiset; it is therefore even in z and real for real z. □

2.3. **The characteristic coefficients.** On the zero-twist axis $c(0,\tau) = 0$. For real nonzero small θ, the scalar zero exclusion of DGRS Lemma 7.4 implies $c \neq 0$. The horizontal operators $D = d_z - \alpha_\tau \wedge L$ and $W = d\alpha_\tau \wedge$ satisfy

$$[B, D] = 0, \qquad D^2 = WB. \tag{25}$$

If f is a nonzero scalar cut state with $Bf = cf$ and $c \neq 0$, then

$$D^2 f = cWf \neq 0, \qquad B(Df) = cDf.$$

Thus $Df \neq 0$ and c is also a degree-one eigenvalue. Therefore

$$\det(s\,\mathrm{Id} - A_1) = (s-c)(s-t). \tag{26}$$

The equality extends first to complex z by holomorphy and then to $z = 0$ by continuity. From the primary first variation,

$$t(0,\tau) = r\tau + o(\tau), \qquad r > 0. \tag{27}$$

Only a derivative on this untwisted axis is used.

Lemma 2.3. *On a smaller fixed disk there are functions C and T, holomorphic in z and continuous in τ locally uniformly in z, such that*

$$c(z,\tau) = z^2 C(z,\tau), \qquad t(z,\tau) = t(0,\tau) + z^2 T(z,\tau). \tag{28}$$

Their values at the origin are

$$C(0,0) = T(0,0) = -\kappa/2. \tag{29}$$

In particular $c(z,\tau) \neq 0$ for sufficiently small complex $z \neq 0$.

Proof. If h is even and holomorphic, with $h(0,\tau) = 0$, choose a radius $0 < a < \rho$. Its quotient by z^2 has the Cauchy representation

$$q_h(z,\tau) := \frac{h(z,\tau)}{z^2} = \frac{1}{2\pi i}\int_{|w|=a} \frac{h(w,\tau)}{w^2(w-z)}\,dw, \qquad |z| < a, \tag{30}$$

where the value at $z = 0$ is removable. For a smaller radius $0 < a' < a$,

$$\sup_{|z|\le a'} |q_h(z,\tau) - q_h(z,\tau')| \le \frac{1}{a(a-a')} \sup_{|w|=a} |h(w,\tau) - h(w,\tau')|. \tag{31}$$

Lemma 2.1 makes the right side tend to zero as $\tau' \to \tau$. Apply this to $h = c$ and $h = t - t(0,\tau)$. Evenness comes from Lemma 2.2, and both functions vanish at $z = 0$. The hyperbolic expansion (13) gives (29). Shrinking the neighborhood once more, we may arrange

$$|C(z,\tau) + \kappa/2| < \kappa/4, \qquad |C(z,\tau)| \ge \kappa/4,$$

which proves the scalar nonvanishing assertion. □

Whenever A_1 is invertible, $D : \mathcal{U}^1_{z,\tau} \to \mathcal{U}^2_{z,\tau}$ is injective, because

$$Du = 0 \implies D^2 u = WBu = 0 \implies Bu = 0 \implies u = 0.$$

Here the middle implication uses the contact Lefschetz isomorphism $W : \mathcal{U}^1_{z,\tau} \to \mathcal{U}^3_{z,\tau}$. Since $[B, D] = 0$, the image of D is a B-invariant subspace. Thus (26) divides $\det(s\,\mathrm{Id} - A_2)$ wherever A_1 is invertible. At every parameter, let Q be the monic quadratic quotient and $\mathcal{R}$ the remainder in the polynomial division

$$P_2(s) = P_1(s)Q(s) + \mathcal{R}(s), \qquad \deg \mathcal{R} < 2.$$

The coefficients of Q and $\mathcal{R}$ inherit holomorphy in z and continuity in τ. For each fixed small $\tau > 0$, $t(0,\tau) \neq 0$, so its holomorphic z-zeros are discrete unless it is identically zero, which it is not. Lemma 2.3 gives $c(z,\tau) \neq 0$ for $z \neq 0$; at $\tau = 0$ the hyperbolic identity (12) applies. Thus the invertible locus is dense, and the remainder vanishes identically. Denoting the two coefficients of Q by $e_1(z,\tau)$ and $e_2(z,\tau)$, we obtain

$$\det(s\,\mathrm{Id} - A_2) = (s-c)(s-t)Q(s), \qquad Q(s) = s^2 - e_1 s + e_2. \tag{32}$$

Monic division preserves holomorphy in z and continuity in τ. Write $c_0(z) := c(z,0)$ for the holomorphic continuation of the real hyperbolic root. The identities in (12) extend from real characters by holomorphy and give

$$e_1(z,0) = 0, \qquad e_2(z,0) = -c_0(z)^2. \tag{33}$$

On the zero-twist axis CDDP Theorem 3 gives horizontal zero multiplicities one in degree one and three in degree two, for all small $\tau > 0$. The additional middle root is the nonzero $t(0,\tau)$ carried by D. Consequently

$$e_1(0,\tau) = e_2(0,\tau) = 0. \tag{34}$$

Lemma 2.4 (The relative estimate)**.** *There is a function A, holomorphic in z and continuous in τ locally uniformly on a fixed disk, such that*

$$c = z^2 C, \qquad t = t(0,\tau) + z^2 T, \qquad e_1 = z^2 A. \tag{35}$$

The three quotient values satisfy

$$C(0,0) = T(0,0) = -\kappa/2, \qquad A(0,0) = 0. \tag{36}$$

Proof. The first two factors have already been constructed in Lemma 2.3, before the polynomial division. The coefficient e_1 is now defined by (32); it is even and holomorphic in z, and $e_1(0,\tau) = 0$ by (34). Apply (30)–(31) with $h = e_1$. The identity $e_1(z,0) = 0$ implies $A(z,0) = 0$, including its removable value at the origin. This proves $A(0,0) = 0$. □

2.4. **The crossing and the full zero complex.** Choose $K > 0$ such that $|T| \le K$. On a sufficiently small positive metric interval,

$$t(0,\tau) \ge \tfrac{r}{2}\tau, \qquad t(\theta,\tau) = t(0,\tau) + \theta^2 T(\theta,\tau).$$

For $\tau_1 = 4K\theta^2/r$,

$$t(\theta,\tau_1) \ge \tfrac{r}{2}\tau_1 - K\theta^2 = K\theta^2 > 0, \qquad t(\theta,0) = c_0(\theta) < 0.$$

Continuity supplies a root

$$t(\theta,\sigma_\theta) = 0, \qquad 0 < \sigma_\theta \le 4K\theta^2/r. \tag{37}$$

More generally, every root in the chosen positive interval satisfies

$$\tfrac{r}{2}\sigma_\theta \le \theta^2 |T(\theta,\sigma_\theta)| \le K\theta^2.$$

Writing $t(0,\tau) = \tau(r + \eta(\tau))$, where $\eta(\tau) \to 0$, the root equation becomes

$$\frac{\sigma_\theta}{\theta^2} = -\frac{T(\theta,\sigma_\theta)}{r + \eta(\sigma_\theta)} \longrightarrow \frac{\kappa}{2r}. \tag{38}$$

No selection with uniqueness or monotonicity is needed.

At a crossing $c \neq 0$, so the degree-one zero eigenvalue is simple.

For completeness, off the zero set of t both scalar and horizontal one-zero spaces vanish. DGRS Proposition 7.3, using pullback acyclicity and contact splitting, excludes the full zero complex. DGRS Theorem 2 then makes the ordinary zeta value constant on each such connected interval. The proof below does not differentiate that ordinary value or take its limit through a crossing.

Fix a crossing (θ,σ_θ) and suppress its parameters. Let V^k be the horizontal generalized zero space and put $m = \dim V^2$. Scalar exclusion and contact Lefschetz give

$$(\dim V^0, \dots, \dim V^4) = (0,1,m,1,0).$$

The degree-one zero is simple because its other cut eigenvalue is $c \neq 0$. In particular $L|_{V^1} = 0$; Lefschetz then gives $L|_{V^3} = 0$ as well. Write $\alpha = \alpha_{\sigma_\theta}$ and use left wedging $jv = \alpha \wedge v$. The full generalized zero complex has spaces $C^k = V^k \oplus jV^{k-1}$, where $V^{-1} = V^5 = 0$. Its dimensions and differential are

$$(\dim C^0, \dots, \dim C^5) = (0,1,m+1,m+1,1,0), \qquad d_\theta(u+jv) = Du + Wv + j(Lu - Dv). \tag{39}$$

The resonant de Rham theorem makes this complex exact. Write d_k for the restriction of d_θ to C^k and D_k for the restriction of the horizontal differential D to V^k. In degree one, $C^1 = V^1$ and

$$d_1 u = (Du, 0), \qquad \ker d_1 = 0,$$

so $D_1 : V^1 \to V^2$ is injective. In degree three, $WV^2 \subset V^4 = 0$, $DV^3 \subset V^4 = 0$, and $L|_{V^3} = 0$, giving

$$d_3(u + jv) = -jD_2v.$$

Exactness in degree four, where $C^4 = jV^3$ and $C^5 = 0$, therefore makes $D_2 : V^2 \to V^3$ surjective. Finally,

$$D_2D_1 = WB|_{V^1} = 0, \qquad 1 = \dim \operatorname{im} D_1 \le \dim \ker D_2 = m - 1,$$

so $m \ge 2$. The nonzero root c occupies one dimension of the rank-four middle cut, giving $m \le 3$. Since

$$P_2(s) = (s - c)sQ(s), \qquad Q(s) = s^2 - e_1s + e_2,$$

the order of $s = 0$ in P_2 is exactly m, and hence

$$e_2 = 0, \qquad m = 3 \iff e_1 = 0, \qquad m = 2 \iff e_1 \neq 0. \tag{40}$$

This argument uses exactness of the full complex; it does not assume exactness of the horizontal sequence.

3. Fixed-cut torsion and the ordinary zeta value

3.1. **The ordinary divisor.** Let $G(s, z, \tau)$ denote the complementary factor obtained by removing the fixed spectral cut, in the convention of [26, (6.4)–(6.5)]. For each parameter it is holomorphic and nonvanishing on the closed spectral disk. The full cut in degree k is $\mathcal{U}^k \oplus \alpha \wedge \mathcal{U}^{k-1}$, so its determinant is P_kP_{k-1}, with $P_{-1} = P_5 = 1$. The weighted full-cut determinant telescopes:

$$\begin{aligned}\mathfrak{D}_{\mathrm{cut}}(s) &= \prod_{k=0}^{5} (P_kP_{k-1})^{(-1)^k k} = \prod_{k=0}^{4} P_k^{(-1)^{k+1}} \\ &= \frac{P_1P_3}{P_0P_2P_4} = \frac{P_1^2}{P_0^2P_2} = \frac{s - t}{(s - c)Q(s)}.\end{aligned} \tag{41}$$

The middle equality uses the contact Lefschetz identities $P_3 = P_1$ and $P_4 = P_0$. In flow dimension five the convention of [26, (6.5)] is $G = Z\mathfrak{D}_{\mathrm{cut}}$. Thus the ordinary zeta divisor is

$$Z(s, z, \tau) = G(s, z, \tau)\frac{(s - c)(s^2 - e_1s + e_2)}{s - t}. \tag{42}$$

Only the following relative normalization is needed; no joint ordinary zeta evaluation at $s = 0$ is assumed.

3.2. **Variation on the full cut.** A dot denotes differentiation with respect to the metric parameter τ, and ι_Y denotes contraction by a vector field Y. For a degree-preserving endomorphism A of a graded finite complex $C^\bullet$, its supertrace is

$$\operatorname{str}_{C^\bullet} A = \sum_k (-1)^k \operatorname{tr}_{C^k} A.$$

Lemma 3.1 (Arbitrary full cut variation)**.** *For fixed real nonzero small θ, put $C^\bullet_{\theta,\tau} = \operatorname{ran} \Pi_{\theta,\tau}$ for the full cut (11). Define*

$$H(\theta, \tau) = \operatorname{str}_{C^\bullet_{\theta,\tau}}(\Pi_{\theta,\tau}\alpha_\tau \wedge \iota_{\dot X_\tau}).$$

Then

$$\frac{G(0, \theta, \tau)}{G(0, \theta, 0)} = \exp\left(\int_0^\tau H(\theta, v)\, dv\right). \tag{43}$$

The finite trace H has a bounded continuous extension to $\theta = 0$. Consequently the left side tends to one uniformly as $\tau \to 0$ and $\theta \to 0$ through real nonzero characters.

Proof. Hold θ fixed and abbreviate $C^\bullet_\tau = C^\bullet_{\theta,\tau}$ and $U^k_\tau = \mathcal{U}^k_{\theta,\tau}$. When a metric subscript is omitted, all operators are evaluated at the current parameter: in particular $L = L_{\theta,\tau}$, $X = X_\tau$, $\alpha = \alpha_\tau$, and $\Pi = \Pi_{\theta,\tau}$; also write $U^k = U^k_\tau$. Write $W_\tau = d\alpha_\tau\wedge$. The right-wedge contact splitting and the Lefschetz map are

$$C^k_\tau = U^{k-1}_\tau \wedge \alpha_\tau \oplus U^k_\tau, \qquad \ell_\tau u = u \wedge d\alpha_\tau, \qquad U^{-1}_\tau = U^5_\tau = 0.$$

We abbreviate $\ell = \ell_\tau$ at the current parameter. For $f \in U^{k-1}$ and $g \in U^k$, the contact chirality Γ_τ is the degree-reversing involution given, for $k = 0, 1, 2$, by the following formula, abbreviating $\Gamma = \Gamma_\tau$:

$$\Gamma(f \wedge \alpha + g) = \ell^{2-k} g \wedge \alpha + \ell^{3-k} f; \tag{44}$$

in the higher degrees its action is the inverse. Here powers and inverse powers of ℓ mean the appropriate contact Lefschetz maps between horizontal degrees.

The cut complex is acyclic for every real nonzero small θ. Its zero part computes $H^\bullet(M; d_\theta) = 0$. On a nonzero resonance part, $\iota_X L^{-1}$ is a cochain contraction by Cartan's identity. The fixed cut commutes with d_θ, ι_{X_τ}, wedging by α_τ, and W_τ; the smooth inverse Lefschetz maps commute as well. Thus the contact chirality Γ_τ preserves the cut. These statements hold on every generalized resonance, not only on a zero block.

Chaubet–Dang initially choose a cut containing only zero at the reference parameter. We give the trace calculation on the present full cut. The smooth-family projector regularity of [26, Proposition 7.2 and Appendix B] supplies differentiable finite cut matrices in τ, with a loss of anisotropic levels. This fixed-character differentiability is used only for the torsion variation formula.

For an acyclic finite complex $C^\bullet$ with chirality Γ, $\tau(C^\bullet, \Gamma)$ denotes the nonzero scalar torsion obtained from its chirality determinant element under the canonical map $\det C^\bullet \to \det H^\bullet(C) = \mathbb{C}$, with the convention of [26]. This is the finite-complex torsion, distinct from the analytic torsion $T_\chi(\Sigma)$. Here $\det C^\bullet = \bigotimes_k (\det C^k)^{(-1)^k}$, where the exponent -1 denotes the dual line.

Choose a reference time τ_*, write $C = C^\bullet_{\theta,\tau_*}$, and define the cochain transport $S_\tau : C \to C^\bullet_\tau$ and the transported chirality $\widetilde{\Gamma}_\tau$ by

$$S_\tau = \Pi_{\theta,\tau}|_C, \qquad \widetilde{\Gamma}_\tau = S_\tau^{-1} \Gamma_\tau S_\tau.$$

The maps S_τ are locally invertible cochain maps because $d_\theta \Pi_{\theta,\tau} = \Pi_{\theta,\tau} d_\theta$. Thus the transported differential is constant:

$$S_\tau^{-1} d_\theta S_\tau = d_\theta|_C. \tag{45}$$

At the reference time,

$$\Pi\dot{\Pi}\Pi = 0, \qquad [\Pi, \Gamma] = 0, \qquad \dot{\widetilde{\Gamma}} = \Pi\dot{\Gamma}|_C.$$

For this fixed acyclic complex the chirality determinant element therefore gives

$$\partial_\tau \log \tau(C^\bullet_\tau, \Gamma_\tau) = \tfrac{1}{2} \operatorname{str}_C(\dot{\widetilde{\Gamma}}\widetilde{\Gamma}) = \sum_{k=0}^{2} (-1)^{k+1} \operatorname{tr}_{C^k}(\Pi\Gamma\dot{\Gamma}). \tag{46}$$

To evaluate these traces, define the scalar h, the horizontal two-form δ, and the bundle maps A_j on horizontal degree j for $j = 0, 1$ by

$$\begin{aligned} h = \alpha(\dot{X}) = -\dot{\alpha}(X), \qquad \delta = d\dot{\alpha} + (\iota_X d\dot{\alpha}) \wedge \alpha, \\ \mathsf{A}_0 u = 2(\ell^2)^{-1}(u \wedge \delta \wedge d\alpha), \qquad \mathsf{A}_1 u = \ell^{-1}(u \wedge \delta). \end{aligned} \tag{47}$$

Here δ is horizontal. Differentiating (44) gives the following diagonal blocks of $\Gamma\dot\Gamma$:

k	$U^{k-1}\wedge\alpha$	U^k
0	0	A_0-h
1	A_0+h	A_1-h
2	A_1+h	$-h$

The remaining blocks interchange the summands and have zero trace after compression by Π. Define

$$a_j=\operatorname{tr}_{U^j}(\Pi\mathsf{A}_j)\quad(j=0,1),\qquad h_j=\operatorname{tr}_{U^j}(\Pi h)\quad(0\le j\le 4).$$

Writing $\gamma_k=\operatorname{tr}_{C^k}(\Pi\Gamma\dot\Gamma)$, we obtain

$$\begin{aligned}\gamma_0&=a_0-h_0,\\ \gamma_1&=a_0+h_0+a_1-h_1,\\ \gamma_2&=a_1+h_1-h_2,\\ -\gamma_0+\gamma_1-\gamma_2&=2h_0-2h_1+h_2.\end{aligned}\tag{48}$$

On the other hand, the diagonal of $\alpha\wedge\iota_{\dot X}$ is multiplication by h on $U^{k-1}\wedge\alpha$ and zero on U^k. Thus

$$H=\operatorname{str}_C(\Pi\alpha\wedge\iota_{\dot X})=-h_0+h_1-h_2+h_3-h_4.\tag{49}$$

The cut Lefschetz isomorphisms intertwine both Π and multiplication by h, so $h_3=h_1$ and $h_4=h_0$. Combining (46)–(49) proves

$$\partial_\tau\log\tau(C^\bullet_\tau,\Gamma_\tau)=-H.\tag{50}$$

The computation requires neither a nilpotent compressed operator nor a cut containing only the zero resonance.

Put $q=(\dim M-1)/2=2$. For the full ranks in (11), the parity exponent Q_{cut} in the sign of [26, (6.6)] is constant:

$$Q_{\mathrm{cut}}=\sum_{k=0}^{2}(-1)^k(3-k)\dim C^k=3-6+6=3.$$

The finite-cut formula [26, (6.6)], with $q=2$, reads

$$\tau_{\mathrm{dyn}}(\theta,\tau)=(-1)^{Q_{\mathrm{cut}}}G(0,\theta,\tau)\tau(C^\bullet_{\theta,\tau},\Gamma_\tau).\tag{51}$$

For the fixed acyclic character, contact deformation invariance and (50) therefore give

$$0=\partial_\tau\log\tau_{\mathrm{dyn}}=\partial_\tau\log G(0,\theta,\tau)-H(\theta,\tau),$$

and hence

$$\frac{G(0,\theta,\tau)}{G(0,\theta,0)}=\exp\!\left(\int_0^\tau H(\theta,v)\,dv\right).$$

Set $\mathcal{K}_\tau=\alpha_\tau\wedge\iota_{\dot X_\tau}$. Using the fixed sources f_j and dual tests ℓ_i of (20), define in each full degree

$$F_{ij}(z,\tau)=\ell_i(\Pi_{z,\tau}f_j),\qquad J_{ij}(z,\tau)=\ell_i(\Pi_{z,\tau}\mathcal{K}_\tau\Pi_{z,\tau}f_j).$$

Write F_k,J_k for these matrices in degree k. The coordinate matrix of the compressed bundle map is $F^{-1}J$. Its continuity follows from the sequence of mappings

$$\mathcal{H}_2\xrightarrow{\Pi_{z,\tau}}\mathcal{H}_1\xrightarrow{\mathcal{K}_\tau}\mathcal{H}_1\xrightarrow{\Pi_{z,\tau}}\mathcal{H}_0\xrightarrow{\ell_i}\mathbb{C}.$$

The matrices F remain uniformly invertible. Define the extension of the finite trace to complex z by

$$H(z,\tau)=\sum_{k=0}^{5}(-1)^k\operatorname{tr}\bigl(F_k(z,\tau)^{-1}J_k(z,\tau)\bigr)$$

This extends continuously across $z = 0$. Choose a uniform bound $K_H > 0$ with $|H| \le K_H$. Then

$$\left|\frac{G(0,\theta,\tau)}{G(0,\theta,0)} - 1\right| \le \exp\left(\int_0^{|\tau|} |H(\theta,v)|\,dv\right) - 1 \le e^{K_H|\tau|} - 1.$$

Only the finite trace is extended to the nonacyclic character; the scalar acyclic torsion identity has been used at $\theta \neq 0$. □

3.3. **Specialization at the crossing.**

Theorem 3.2. *For sufficiently small real $\theta \neq 0$, choose any crossing σ_θ furnished by (37). At that metric, either $Z(s,\theta,\sigma_\theta)$ has a simple zero at $s = 0$, or it is regular and nonzero there and*

$$\frac{Z(0,\theta,\sigma_\theta)}{Z(0,\theta,0)} \longrightarrow 0. \tag{52}$$

In either case the ordinary assertion of Conjecture 9 fails at every sufficiently small such crossing. The geometric inputs are detailed in the appendix, with explicit attribution to the primary spectral formulas used there.

Proof. At the hyperbolic endpoint (12), (33), and (42) give the exact value

$$C_\theta := Z(0,\theta,0) = -G(0,\theta,0)c_0(\theta)^2 \neq 0. \tag{53}$$

At a crossing, (40) gives $e_2 = 0$, so $Q(s) = s(s - e_1)$. In the formulas at this crossing, c, e_1, e_2, and Q are evaluated at (θ,σ_θ) when their parameter arguments are omitted. Fix (θ,σ_θ) first and cancel the common s factor in (42). In both multiplicity cases this gives

$$\begin{aligned} Z(s,\theta,\sigma_\theta) &= G(s,\theta,\sigma_\theta)(s-c)(s-e_1), \\ Z(0,\theta,\sigma_\theta) &= G(0,\theta,\sigma_\theta)ce_1. \end{aligned} \tag{54}$$

If $m = 3$, then $e_1 = 0$ and the first line is $G(s)(s-c)s$. Its zero at the origin is simple because $c \neq 0$ and $G(0) \neq 0$. If $m = 2$, then $e_1 \neq 0$ and the value in the second line is nonzero.

Divide the second line of (54) by (53):

$$\frac{Z(0,\theta,\sigma_\theta)}{C_\theta} = -\frac{G(0,\theta,\sigma_\theta)}{G(0,\theta,0)}\frac{c(\theta,\sigma_\theta)}{c_0(\theta)}\frac{e_1(\theta,\sigma_\theta)}{c_0(\theta)}. \tag{55}$$

This identity includes the simple-zero case. The first factor tends to one by Lemma 3.1 and $\sigma_\theta = O(\theta^2)$. The other two limits follow directly from the quotient estimates:

$$\frac{c(\theta,\sigma_\theta)}{c_0(\theta)} = \frac{C(\theta,\sigma_\theta)}{C(\theta,0)} \longrightarrow 1, \qquad \frac{e_1(\theta,\sigma_\theta)}{c_0(\theta)} = \frac{A(\theta,\sigma_\theta)}{C(\theta,0)} \longrightarrow 0. \tag{56}$$

Here $C(\theta,0) \to -\kappa/2 \neq 0$ and $A(\theta,\sigma_\theta) \to A(0,0) = 0$. This proves (52) and the three estimates in (10).

Fried's hyperbolic theorem [6] identifies $|C_\theta| = T_{\chi_\theta}(\Sigma, g_0)^2$. Acyclic unitary analytic torsion is independent of the metric on this odd-dimensional base, so this is also the target at g_{σ_θ}. A simple zero contradicts the required nonvanishing; in the second case the modulus ratio is less than one for all sufficiently small θ, contradicting the required value. The full zero complex, rather than horizontal exactness or semisimplicity, supplied the exhaustive two cases. □

Proof of Theorem 1.1. Use the base constructed in the appendix and any sequence of nonzero real $\theta_j \to 0$. Choose the crossings $\sigma_{\theta_j} = O(\theta_j^2)$ from (37), and put $g_j = g_{\sigma_{\theta_j}}$, $\chi_j = \chi_{\theta_j}$. In both cases of (40), $Q(s) = s(s - e_1)$ at the crossing, so (42) is the holomorphic product $G(s)(s-c)(s-e_1)$. Its value is $G(0)ce_1$, including the case $e_1 = 0$. Equation (55) therefore holds in both cases. The three factor limits in Theorem 3.2, and the hyperbolic torsion identity, give the assertion. Smooth convergence of the metrics follows from $\sigma_{\theta_j} \to 0$. □

Remark 3.3 (Why a fixed complex disk is essential). The quantitative ingredient is uniform holomorphy on a fixed complex character disk. Real C^0 continuity, evenness, the two vanishing axes, a real crossing, and divisor cancellation do not give the relative estimate. A concrete coefficient model on $\tau \geq 0$ is

$$a = \theta^2, \quad c = -a, \quad t = \tau - a, \quad e_1 = \frac{2a\tau}{\tau + a}, \quad e_2 = a(\tau - a), \quad G = 1,$$

with $e_1(0,0) = 0$. It has the required axis identities, is continuous and even in real θ, and has its crossing at $\tau = a$. For $a > 0$ and $t \neq 0$, the divisor gives

$$Z(0, \theta, \tau) = \frac{ce_2}{t} = -a^2.$$

At $\tau = a$ one has $e_1 = a$ and $e_2 = 0$, so the specialized product is

$$Z(s, \theta, a) = (s + a)(s - a), \qquad Z(0, \theta, a) = -a^2.$$

Thus the crossing alone forces no ordinary value discrepancy. The coefficient is holomorphic in θ at each fixed $\tau > 0$, but its poles $\theta = \pm i\sqrt{\tau}$ approach the origin. It fails exactly the uniform-disk hypothesis furnished by (17) for the actual geometric family.

The present proof retains all geometric spectral inputs but replaces mixed metric-character differentiability by uniform complex-character holomorphy and a finite-cut normalization ratio. It neither exchanges evaluation at zero with meromorphic parameter specialization nor assumes that ordinary zeta values are continuous through a crossing.

Appendix A. Geometric and spectral inputs

A.1. **The hyperbolic base and acyclic characters.** We first construct the base and the acyclic character interval, then give the conformal and spectral inputs in their source conventions.

Let S be a closed oriented surface of genus two. In a symplectic basis of its first integral homology $H_1(S; \mathbb{Z})$, define the matrices A, T, and the intersection-form matrix J_{sp} as follows. Here Id_m denotes the $m \times m$ identity matrix, and a superscript T denotes matrix transpose:

$$A = \begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix}, \qquad T = \begin{pmatrix} A & -\mathrm{Id}_2 \\ \mathrm{Id}_2 & 0 \end{pmatrix}, \qquad J_{\mathrm{sp}} = \begin{pmatrix} 0 & \mathrm{Id}_2 \\ -\mathrm{Id}_2 & 0 \end{pmatrix}. \tag{57}$$

Since $A^T = A$, a block multiplication gives

$$T^T J_{\mathrm{sp}} T = \begin{pmatrix} A & \mathrm{Id}_2 \\ -\mathrm{Id}_2 & 0 \end{pmatrix} \begin{pmatrix} \mathrm{Id}_2 & 0 \\ -A & \mathrm{Id}_2 \end{pmatrix} = J_{\mathrm{sp}}. \tag{58}$$

Thus $T \in \mathrm{Sp}(4, \mathbb{Z})$, the group of integral matrices preserving J_{sp}. Its characteristic polynomial p is

$$\begin{aligned} p(x) &= \det(x\,\mathrm{Id}_4 - T) = \det(x^2\,\mathrm{Id}_2 - xA + \mathrm{Id}_2) \\ &= (x^2 - 2x + 1)(x^2 + x + 1) - x^2 = x^4 - x^3 - x^2 - x + 1. \end{aligned} \tag{59}$$

Let $\overline{p}$ denote reduction of p modulo two, let Φ_5 denote the fifth cyclotomic polynomial, and let $\mathrm{ord}_G(a)$ denote the multiplicative order of a in a group G. Then

$$\overline{p}(x) = x^4 + x^3 + x^2 + x + 1 = \Phi_5(x), \qquad \mathrm{ord}_{(\mathbb{Z}/5\mathbb{Z})^\times}(2) = 4. \tag{60}$$

Consequently $\overline{p}$ is irreducible over $\mathbb{F}_2$, and p is irreducible over $\mathbb{Q}$. Moreover,

$$p(1) = -1, \qquad p(2) = 3. \tag{61}$$

The resulting real root in $(1, 2)$ excludes cyclotomicity. The nonzero coefficient of x excludes a representation $p(x) = q(x^k)$ for any polynomial q and integer $k > 1$.

The mapping-class action onto $\mathrm{Sp}(4,\mathbb{Z})$ is surjective. The homological criterion of Casson–Bleiler, in the form of [33, Propositions 1–2], therefore supplies an orientation-preserving pseudo-Anosov $\varphi : S \to S$ inducing T. Its mapping torus

$$\Sigma = (S \times [0,1])/((x,1) \sim (\varphi(x),0)) \tag{62}$$

is a closed oriented hyperbolic three-manifold by fibered hyperbolization [34]; choose a hyperbolic metric g_0 on it. The homology Wang sequence, using $\varphi_*|_{H_0(S;\mathbb{R})} = \mathrm{Id}$, gives

$$0 \longrightarrow \mathrm{coker}(T - \mathrm{Id}_4) \longrightarrow H_1(\Sigma;\mathbb{R}) \longrightarrow \mathbb{R} \longrightarrow 0. \tag{63}$$

Here the cokernel is taken on $H_1(S;\mathbb{R})$. By (61), $\det(\mathrm{Id}_4 - T) = p(1) \neq 0$, so the cokernel is zero. Since an endomorphism of a finite-dimensional space has equal kernel and cokernel dimensions, we obtain

$$b_1(\Sigma) = 1 + \dim\ker(T - \mathrm{Id}_4) = 1. \tag{64}$$

Fix a nonzero real harmonic one-form ω on (Σ, g_0). For real θ, let d_θ^* be the $L^2(g_0)$ adjoint of the twisted differential d_θ, and define the degree-k Hodge Laplacian by

$$d_\theta = d + i\theta\omega\wedge, \qquad \Delta_{\mathrm{H},\theta}^{(k)} = d_\theta^* d_\theta + d_\theta d_\theta^* \quad \text{on } k\text{-forms}. \tag{65}$$

The connection is unitary when $\theta \in \mathbb{R}$. Choose a loop γ with $I_\gamma = \int_\gamma \omega \neq 0$. If

$$0 < |\theta| < \frac{2\pi}{|I_\gamma|}, \tag{66}$$

then $e^{i\theta I_\gamma} \neq 1$. Changing the convention for parallel transport replaces θ by $-\theta$ and does not affect this nontriviality argument.

Write $b_k = \dim H^k(\Sigma;\mathbb{R})$ for the degree-k Betti number. At $\theta = 0$, the kernels of the scalar and one-form Hodge Laplacians have dimensions $b_0 = b_1 = 1$. Choose $\eta > 0$ below their first positive eigenvalues, and write $\mathbf{1}_{[0,\eta)}(D)$ for the spectral projection of a self-adjoint operator D onto that interval. Smooth elliptic perturbation theory, after shrinking η and the character interval if necessary, gives

$$\mathrm{rank}\,\mathbf{1}_{[0,\eta)}(\Delta_{\mathrm{H},\theta}^{(0)}) = \mathrm{rank}\,\mathbf{1}_{[0,\eta)}(\Delta_{\mathrm{H},\theta}^{(1)}) = 1. \tag{67}$$

Write $\mu_0(\theta)$ for the eigenvalue in the scalar slot, and choose an eigenfunction f_θ with $\|f_\theta\|_2 = 1$, where $\|\cdot\|_2$ is the $L^2(g_0)$ norm on the base. A scalar harmonic section is parallel. Nontrivial holonomy therefore implies

$$\ker \Delta_{\mathrm{H},\theta}^{(0)} = 0, \qquad 0 < \mu_0(\theta) < \eta \quad (0 < |\theta| \ll 1). \tag{68}$$

The differential commutes with the Hodge Laplacians, so

$$\begin{aligned} \|d_\theta f_\theta\|_2^2 &= \mu_0(\theta) > 0, \\ \Delta_{\mathrm{H},\theta}^{(1)} d_\theta f_\theta &= d_\theta \Delta_{\mathrm{H},\theta}^{(0)} f_\theta = \mu_0(\theta) d_\theta f_\theta. \end{aligned} \tag{69}$$

This nonzero exact form fills the unique small one-form slot in (67). In particular,

$$H^0(\Sigma;\chi_\theta) = H^1(\Sigma;\chi_\theta) = 0. \tag{70}$$

Unitary Hodge duality, or Poincaré duality applied also to the inverse character $\chi_{-\theta}$, gives

$$H^k(\Sigma;\chi_\theta) = 0 \quad (0 \le k \le 3), \qquad 0 < |\theta| \ll 1. \tag{71}$$

The same argument gives the gap needed for the weight-one Selberg factor. When $\theta \neq 0$ is small, acyclicity identifies coclosed one-forms with the coexact sector. Every one-form eigenvalue below η is exact. Thus

$$\mathrm{Spec}\Big(\Delta_{\mathrm{H},\theta}^{(1)}\big|_{\ker d_\theta^*}\Big) \subset [\eta,\infty), \qquad 0 < |\theta| \ll 1. \tag{72}$$

This is a gap on one-forms in the indicated sector; the scalar eigenvalue in (68) tends to zero.

For completeness, the quadratic coefficient of that scalar eigenvalue can be read directly from the scalar operator. Let $V = \mathrm{vol}_{g_0}(\Sigma)$ and $f_0 = V^{-1/2}$. Harmonicity gives $d^*\omega = 0$. Write K_1, K_2 for the linear and quadratic coefficients in the scalar-operator expansion:

$$\Delta^{(0)}_{\mathrm{H},\theta} = \Delta^{(0)}_{\mathrm{H},0} + \theta K_1 + \theta^2 K_2, \qquad K_1 f = -2i\langle \omega, df\rangle_{g_0}, \qquad K_2 f = |\omega|^2 f. \tag{73}$$

In particular $K_1 f_0 = 0$. Primes in the next two equations denote θ-derivatives at $\theta = 0$. Differentiating the simple eigenvalue equation at zero gives

$$\mu_0'(0) = \langle f_0, K_1 f_0\rangle = 0, \qquad \Delta^{(0)}_{\mathrm{H},0} f_0' = 0. \tag{74}$$

Choose the phase of f_θ so that $\langle f_0', f_0\rangle = 0$. Then $f_0' = 0$, and the second derivative gives

$$\frac{1}{2}\mu_0''(0) = \langle f_0, K_2 f_0\rangle = \frac{\|\omega\|_2^2}{V} =: \kappa > 0. \tag{75}$$

Complex conjugation identifies the scalar operators at θ and $-\theta$. The simple eigenvalue is therefore even, and analytic elliptic perturbation theory yields

$$\mu_0(\theta) = \kappa\theta^2 + O(\theta^4), \qquad -1 + \sqrt{1 - \mu_0(\theta)} = -\frac{\kappa}{2}\theta^2 + O(\theta^4). \tag{76}$$

Finally, for the unit sphere bundle $\pi : M \to \Sigma$, the fiber is S^2. Let $E_2^{p,q}$ be the second page of its local-coefficient Leray–Serre spectral sequence, with p the base degree and q the fiber degree. It has

$$E_2^{p,q} = H^p(\Sigma; \chi_\theta \otimes H^q(S^2; \mathbb{C})) = 0, \qquad q \in \{0, 2\}, \tag{77}$$

and no other nonzero rows. The orientation of the bundle makes the fiber cohomology local systems trivial. Hence $H^\bullet(M; \pi^*\chi_\theta) = 0$.

A.2. **The nondegenerate conformal direction.** In this subsection Δ_{g_0} denotes the nonpositive scalar Laplacian used by CDDP, and $X = X_0$, $\alpha = \alpha_0$. Write tangent vectors as $\xi = (\xi_H, \xi_V)$ and $\eta = (\eta_H, \eta_V)$ in the Levi–Civita horizontal/vertical splitting. Define the tangent-bundle rotation I and invariant two-form ψ by

$$\mathsf{I}_{(x,v)}(\xi_H, \xi_V) = (v \times \xi_V, v \times \xi_H), \qquad \psi(\xi, \eta) = d\alpha_0(\mathsf{I}\xi, \eta).$$

Here $\times$ is the cross product determined by g_0 and the base orientation; ψ is CDDP's additional invariant two-form. Let $J(x, v) = (x, -v)$ be velocity reversal. Here "outgoing" denotes the resonant cut selected by the anisotropic spaces fixed above, and "incoming" denotes its image under J^*. Define the closed horizontal zero one-forms by $\mathcal{C}_{\mathrm{cl}} = \mathcal{U}^1_{0,0} \cap \ker d$, and set $\mathcal{C}_\psi = \{v \circ \mathsf{I} : v \in \mathcal{C}_{\mathrm{cl}}\}$, where composition means $(v \circ \mathsf{I})(\xi) = v(\mathsf{I}\xi)$. Let π_* denote fiber integration with CDDP's fiber-orientation convention. Normalize a real nonclosed $u \in \mathcal{C}_\psi$ by

$$\pi_*(d\alpha_0 \wedge u) = \omega, \qquad u_* = J^*u, \qquad N = \|\omega\|_2^2. \tag{78}$$

Their Lemma 3.17 and Corollary 5.1 identify the corresponding incoming pushforward with the same ω. Define the distribution F and the smooth function f by

$$\pi_*(\alpha_0 \wedge du \wedge du_*) = F\, d\,\mathrm{vol}_{g_0}, \qquad f = \Delta_{g_0}|\omega|^2. \tag{79}$$

For a smooth function h on Σ, let $\widetilde{h}$ be its invariant lift to the universal cover $\mathbb{H}^3$, and let $d_{\mathbb{H}^3}$ denote hyperbolic distance. CDDP's smoothing operator Q_4 is defined by the hyperbolic-distance kernel

$$\widetilde{Q_4 h}(x) = \int_{\mathbb{H}^3} \frac{\widetilde{h}(y)}{\cosh^4 d_{\mathbb{H}^3}(x, y)}\, d\,\mathrm{vol}_{\mathbb{H}^3}(y). \tag{80}$$

It descends to Σ and extends to distributions [25, §5.1.2]. Their Theorem 5 and Proposition A.1 give

$$Q_4 F = -\frac{1}{6} f, \qquad f \neq 0. \tag{81}$$

Indeed, $f = 0$ would make $|\omega|^2$ constant on the compact base, which is excluded for a nonzero harmonic one-form by the cited proposition. The operator Q_4 has a symmetric periodized smoothing kernel, so it is self-adjoint in the distribution–test-function pairing. Set

$$b = Q_4 f. \tag{82}$$

With $\langle F, b\rangle$ denoting the action of the distribution F on the test function b relative to $d\,\mathrm{vol}_{g_0}$, we have

$$\langle F, b\rangle = \langle Q_4 F, f\rangle = -\frac{1}{6}\|f\|_2^2 \neq 0. \tag{83}$$

The sign in the passage back to M is essential. The orientation induced by CDDP's fiber integration is opposite to the contact orientation $\alpha_0 \wedge (d\alpha_0)^2$. With the latter used for integrals over M, the convention is

$$\int_M \pi^* a\,\Omega = -\int_\Sigma a\,\pi_*\Omega \quad \text{for top forms } \Omega \text{ and smooth } a, \qquad \pi_*(\alpha_0 \wedge (d\alpha_0)^2) = -8\pi\, d\,\mathrm{vol}_{g_0}\,. \tag{84}$$

Equivalently, CDDP's equation (5.6), tested against f, yields

$$S := \int_M \pi^* b\,\alpha_0 \wedge du \wedge du_* = -\langle F, b\rangle = \frac{1}{6}\|f\|_2^2 > 0. \tag{85}$$

The normalization of the outgoing–incoming pairing is equally explicit. For outgoing and incoming horizontal one-forms v, v_*, put

$$\mathcal{B}(v, v_*) = \int_M \alpha_0 \wedge d\alpha_0 \wedge v \wedge v_*. \tag{86}$$

Let $\mathcal{I}u = u \circ \mathsf{I}$ be the induced rotation of one-forms, and let brackets denote de Rham cohomology classes. The identities in [25, Lemma 3.5, (3.57), (3.67), and the calculation preceding (3.59)], together with the normalization (78), give

$$\begin{aligned} u_{\mathrm{cl}} := \mathcal{I}u, \qquad [u_{\mathrm{cl}}] = -\frac{1}{4\pi}[\pi^*\omega], \qquad \pi_*(\psi \wedge u_{\mathrm{cl}}) = \omega, \\ \mathcal{B}(u_{\mathrm{cl}}, J^* u_{\mathrm{cl}}) = -\frac{N}{4\pi}, \qquad \mathcal{B}(u, J^* u) = \frac{N}{4\pi}. \end{aligned} \tag{87}$$

The final sign reversal is (3.57); J commutes with I, so the same rotation is applied to both entries.

The real horizontal zero one-space is the direct sum of one-dimensional closed and nonclosed subspaces. CDDP (3.55)–(3.57) make their pairing matrix diagonal with nonzero diagonal entries. For $g_\tau = e^{-2\tau b} g_0$, set $\beta = \partial_\tau \alpha_\tau|_{\tau=0}$. Primes below denote τ-derivatives at zero on the identified degree-one spectral cut. The subscript cut denotes compression to that cut. Equations (4.22) and (4.38), with $\iota_X \beta = -b \circ \pi$, give

$$\mathcal{B}(P'_{\mathrm{cut}} u, u_*) = iS, \qquad P = -i\mathcal{L}_X, \tag{88}$$

and zero variation on the closed row and column. Consequently, in the closed/nonclosed basis,

$$P'_{\mathrm{cut}} = \begin{pmatrix} 0 & 0 \\ 0 & ir \end{pmatrix}, \qquad B'_{\mathrm{cut}} = -iP'_{\mathrm{cut}} = \begin{pmatrix} 0 & 0 \\ 0 & r \end{pmatrix}, \qquad r = \frac{S}{N/(4\pi)} = \frac{2\pi}{3N}\|f\|_2^2 > 0. \tag{89}$$

This proves the required sign of the companion-root derivative using the cited compressed-variation formula. When $b_1 = 1$, the nonzero entry is the nondegeneracy condition of CDDP's Theorem 3. That theorem and Corollary 4.1 provide the untwisted zero ranks used in the main argument. Small smooth conformal perturbations remain negatively curved.

A.3. **Hyperbolic character clusters.** Here $SO(m)$ is the special orthogonal group in dimension m. Let $\mathbf{1}$ be its trivial representation and let ν_l and σ_p be the representations of $SO(2)$ on $\Lambda^l\mathbb{C}^2$ and on trace-free symmetric tensors of order p, respectively. Thus $\sigma_0 = \mathbf{1}$, $\sigma_1 = \nu_1$, ν_1 is the sum of weights $+1, -1$, and σ_2 is the sum of weights $+2, -2$. Write $Z^D_\nu(a)$ for DGRS's Selberg function with representation ν and complex argument a, suppressing the fixed twist χ_θ. Write $Z^{\mathrm{BO}}_\nu(t)$ for the corresponding Bunke–Olbrich function in the centered argument t. The scalar center in dimension three is $\rho = 1$, so $t = a - 1$ and comparison of the Euler products gives

$$Z^D_\nu(a) = Z^{\mathrm{BO}}_\nu(a-1). \tag{90}$$

The representations relevant to the first three horizontal degrees satisfy

$$\nu_0 = \nu_2 = \mathbf{1}, \qquad \nu_1 \otimes \nu_1 = \mathbf{1}^{\oplus 2} \oplus \sigma_2. \tag{91}$$

Write Λ^j for the exterior-power representation on $\Lambda^j\mathbb{C}^3$, and Sym^2_0 for the trace-free second symmetric power of $\mathbb{C}^3$. The virtual $SO(3)$ lifts, with restriction to $SO(2)$ understood, are

$$\begin{aligned}
&\mathbf{1} \longleftrightarrow \Lambda^0, \\
&\nu_1 \longleftrightarrow \Lambda^1 - \Lambda^0, && \Lambda^1|_{SO(2)} = \mathbf{1} \oplus \nu_1, \\
&\sigma_2 \longleftrightarrow \mathrm{Sym}^2_0 - \Lambda^1, && \mathrm{Sym}^2_0\,|_{SO(2)} = \mathbf{1} \oplus \nu_1 \oplus \sigma_2.
\end{aligned} \tag{92}$$

The subscript 0 denotes trace-free symmetric tensors. Using full symmetric powers in the last line would introduce an extra scalar summand. The pair of weights in each line occurs once. The symmetric Bunke–Olbrich formula is a graded determinant for that pair; it carries no additional square or square root.

Write $A \doteq B$ when A/B extends to a holomorphic nonvanishing function in the spectral neighborhood under discussion. The elliptic determinants below are zeta-regularized in Bunke–Olbrich's convention. Bunke–Olbrich [13, Proposition 4.7], in these conventions, gives

$$\begin{aligned}
Z^D_{\mathbf{1}}(a) &\doteq \det\left(\Delta^{(0)}_{\mathrm{H},\theta} - 1 + (a-1)^2\right), \\
Z^D_{\nu_1}(1+t) &\doteq \frac{\det(\Delta^{(1)}_{\mathrm{H},\theta} + t^2)}{\det(\Delta^{(0)}_{\mathrm{H},\theta} + t^2)}.
\end{aligned} \tag{93}$$

For a small nontrivial real character, d_θ identifies the entire scalar spectrum with the exact one-form spectrum. Thus the exact sector cancels the denominator, including the small eigenvalue $\mu_0(\theta)$. The remaining determinant is on the coexact sector, and (72) excludes zeros or poles for $|t| < \sqrt{\eta}$. At the trivial character, the harmonic contribution is instead

$$\frac{t^{2b_1}}{t^{2b_0}} = t^{2(b_1 - b_0)} = 1. \tag{94}$$

The positive exact spectra still cancel, and the nonharmonic coexact spectrum has a positive gap. This is why the centered weight-one factor contributes no small root, including at $\theta = 0$.

For the weight-two pair, the virtual lift in (92) has Casimir shift $c_{\mathrm{Cas}}(\sigma_2) = 1 - 2^2 = -3$; the subscript distinguishes this constant from the small resonance root c. Let ∇_θ be the Levi–Civita covariant derivative on the indicated tensor bundle tensored with the flat connection d_θ; $\nabla^*_\theta\nabla_\theta$ is its nonnegative rough Laplacian. Denote the relevant Lichnerowicz operators by Δ_2 and Δ_1. In DGRS Appendix B's normalization these and the centered operators are

$$\begin{aligned}
&\Delta_2 = \nabla^*_\theta\nabla_\theta - 6, && \Delta_2 - c_{\mathrm{Cas}}(\sigma_2) = \nabla^*_\theta\nabla_\theta - 3 && \text{on } \mathrm{Sym}^2_0, \\
&\Delta_1 = \nabla^*_\theta\nabla_\theta - 2, && \Delta_1 - c_{\mathrm{Cas}}(\sigma_2) = \nabla^*_\theta\nabla_\theta + 1 && \text{on } \Lambda^1.
\end{aligned} \tag{95}$$

At the DGRS argument $a = s + 2$, the centered parameter is $s + 1$. Hence the numerator and denominator operators are respectively

$$\nabla^*_\theta\nabla_\theta - 2 + 2s + s^2 \quad \text{on } \mathrm{Sym}^2_0, \qquad \nabla^*_\theta\nabla_\theta + 2 + 2s + s^2 \quad \text{on } \Lambda^1. \tag{96}$$

The hyperbolic Bochner inequality used in DGRS's proof of Proposition 7.7 is

$$\nabla_\theta^*\nabla_\theta \geq 3 \quad \text{on } \mathrm{Sym}_0^2. \tag{97}$$

A flat unitary twist introduces no additional curvature term. The rough Laplacian on Λ^1 is nonnegative. Thus the two operators in (96) are invertible whenever

$$|2s + s^2| < \frac{1}{2}, \tag{98}$$

uniformly for the real characters under consideration. The weight-two factor is holomorphic and nonvanishing in this fixed neighborhood.

We now enumerate the factors in DGRS (7.6). In horizontal degree $k \in \{0,1,2\}$, $l \in \{0,\dots,k\}$ indexes the exterior-power splitting, $p \geq 0$ the trace-free symmetric-power degree, and $j \geq 0$ the radial index. The associated Selberg representation is $\nu_l \otimes \nu_{k-l} \otimes \sigma_p$, and its spectral shift h is

$$h = 2(j-l) + p + 2 + k. \tag{99}$$

The complete list with $h \leq 2$ is

k	l	p	j	Selberg factor
0	0	0	0	$Z_{\mathbf{1}}^D(s+2)$
1	1	0	0	$Z_{\nu_1}^D(s+1)$
1	1	1	0	$Z_{\nu_1\otimes\nu_1}^D(s+2)$
2	2	0	0	$Z_{\mathbf{1}}^D(s)$
2	2	0	1	$Z_{\mathbf{1}}^D(s+2)$
2	2	1	0	$Z_{\nu_1}^D(s+1)$
2	2	2	0	$Z_{\sigma_2}^D(s+2)$
2	1	0	0	$Z_{\nu_1\otimes\nu_1}^D(s+2)$

(100)

All remaining shifts satisfy $h \geq 3$. If $|s| < 1/2$, then $\mathrm{Re}(s+h) > 2$, so each remaining factor lies in its zero-free Euler-product half-plane. The normally convergent tail in the cited DGRS factorization is therefore a holomorphic nonvanishing factor. Normal convergence is used here in addition to the individual zero-free property.

Let $\mathscr{D}_k$ denote the individual horizontal dynamical determinant in degree k. Combining (91)–(100) gives

$$\begin{aligned}
\mathscr{D}_0(s) &\doteq Z_{\mathbf{1}}^D(s+2),\\
\mathscr{D}_1(s) &\doteq Z_{\mathbf{1}}^D(s+2)^2,\\
\mathscr{D}_2(s) &\doteq Z_{\mathbf{1}}^D(s)Z_{\mathbf{1}}^D(s+2)^3.
\end{aligned} \tag{101}$$

The degree-two exponent three consists of one scalar factor from $(l,p,j) = (2,0,1)$ and two scalar summands from $(1,0,0)$. The opposite small root comes solely from $(2,0,0)$.

For the scalar eigenvalue $\mu_0(\theta)$, the two shifted scalar equations in (93) are

$$\begin{aligned}
a = s+2: &\qquad \mu_0 + s^2 + 2s = 0, &&\qquad s = -1 \pm \sqrt{1-\mu_0},\\
a = s: &\qquad \mu_0 + s^2 - 2s = 0, &&\qquad s = 1 \pm \sqrt{1-\mu_0}.
\end{aligned} \tag{102}$$

Set

$$c_0(\theta) = -1 + \sqrt{1-\mu_0(\theta)} < 0 \qquad (0 < |\theta| \ll 1). \tag{103}$$

The roots near zero are c_0 and $-c_0$ respectively; the other roots are near -2 and 2. They are simple because the scalar eigenspace has dimension one and $\mu_0(\theta) < 1$. All other scalar eigenvalues remain separated from zero. A fixed sufficiently small circle therefore contains precisely these roots.

DGRS identifies the orders of the individual dynamical determinants with algebraic resonance multiplicities. For the operator $B = -\mathcal{L}_X$, whose spectral coordinate is s, the monic small-cluster characteristic polynomials are consequently

$$P_0(s) = s - c_0, \qquad P_1(s) = (s - c_0)^2, \qquad P_2(s) = (s - c_0)^3(s + c_0). \tag{104}$$

At $\theta = 0$, DGRS Proposition 7.7 gives the same horizontal ranks $(1, 2, 4, 2, 1)$. Contact Lefschetz supplies degrees three and four. No semisimplicity assertion for the degree-two cluster is needed.

The Bunke–Olbrich preprint has a reciprocal inconsistency between its Ruelle Definition 5.1 and the Selberg factorization in Proposition 5.2. We use only its Selberg determinant formula and calibrate the hyperbolic modulus independently by Fried's theorem [6].

The Euler-product check is elementary. Let $\ell(\gamma) = \ell_{g_0}(\gamma)$ be the length of a primitive orbit γ, let $m(\gamma) \in SO(2)$ be its normal rotational holonomy, and write $\chi = \chi_\theta$. For an iterate $j \geq 1$, put

$$q = e^{-j\ell(\gamma)}, \qquad m = m(\gamma)^j \in SO(2). \tag{105}$$

The alternating numerator of the Selberg factorization is

$$1 - q\,\mathrm{tr}(m) + q^2 = \det(\mathrm{Id} - qm). \tag{106}$$

It cancels the Selberg denominator. The remaining logarithm is

$$-\sum_{\gamma \text{ primitive}} \sum_{j \geq 1} \frac{\mathrm{tr}(\chi(\gamma)^j) e^{-sj\ell(\gamma)}}{j} = \log \prod_{\gamma \text{ primitive}} \det(\mathrm{Id} - \chi(\gamma) e^{-s\ell(\gamma)}), \tag{107}$$

the direct Euler product used throughout the article. This normalization issue affects neither (104) nor the spectral reductions above.

References

[1] M. Zworski, *Mathematical study of scattering resonances*, Bull. Math. Sci. 7 (2017), 1–85, DOI, Conjecture 9.

[2] D. B. Ray and I. M. Singer, *R-torsion and the Laplacian on Riemannian manifolds*, Adv. Math. 7 (1971), no. 2, 145–210, doi:10.1016/0001-8708(71)90045-4.

[3] J. Cheeger, *Analytic torsion and the heat equation*, Ann. of Math. (2) 109 (1979), no. 2, 259–322, doi:10.2307/1971113.

[4] W. Müller, *Analytic torsion and R-torsion of Riemannian manifolds*, Adv. Math. 28 (1978), no. 3, 233–305, doi:10.1016/0001-8708(78)90116-0.

[5] W. Müller, *Analytic torsion and R-torsion for unimodular representations*, J. Amer. Math. Soc. 6 (1993), no. 3, 721–753, doi:10.2307/2152781.

[6] D. Fried, *Analytic torsion and closed geodesics on hyperbolic manifolds*, Invent. Math. 84 (1986), no. 3, 523–540, doi:10.1007/BF01388745.

[7] S. Shen, *Analytic torsion and dynamical flow: a survey on the Fried conjecture*, in *Arithmetic L-Functions and Differential Geometric Methods*, Progr. Math. 338, Birkhäuser, Cham, 2021, 247–299, doi:10.1007/978-3-030-65203-6_9.

[8] D. Fried, *Lefschetz formulas for flows*, in *The Lefschetz Centennial Conference, Part III (Mexico City, 1984)*, A. Verjovsky (ed.), Contemp. Math. 58, Part III, Amer. Math. Soc., Providence, RI, 1987, 19–69, doi:10.1090/conm/058.3/893856.

[9] D. Fried, *Meromorphic zeta functions for analytic flows*, Comm. Math. Phys. 174 (1995), no. 1, 161–190, doi:10.1007/BF02099469.

[10] H. Moscovici and R. J. Stanton, *R-torsion and zeta functions for locally symmetric manifolds*, Invent. Math. 105 (1991), no. 1, 185–216, doi:10.1007/BF01232263.

[11] S. Shen, *Analytic torsion, dynamical zeta functions, and the Fried conjecture*, Anal. PDE 11 (2018), no. 1, 1–74, doi:10.2140/apde.2018.11.1, arXiv:1602.00664.

[12] U. Bunke and M. Olbrich, *Selberg zeta and theta functions: a differential operator approach*, Math. Research 83, Akademie-Verlag, Berlin, 1995, ISBN 978-3-05-501690-5.

[13] U. Bunke and M. Olbrich, *Theta and zeta functions for odd-dimensional locally symmetric spaces of rank one*, preprint, 1994, arXiv:dg-ga/9407012, version 1, Proposition 4.7 and its graded-determinant convention.

[14] H. H. Rugh, *Generalized Fredholm determinants and Selberg zeta functions for Axiom A dynamical systems*, Ergodic Theory Dynam. Systems 16 (1996), no. 4, 805–819, doi:10.1017/S0143385700009111.

[15] P. Giulietti, C. Liverani, M. Pollicott, *Anosov flows and dynamical zeta functions*, Ann. of Math. (2) 178 (2013), no. 2, 687–773, Publisher text, doi:10.4007/annals.2013.178.2.6.
[16] P. Giulietti, M. Pollicott, C. Liverani, *Anosov flows and dynamical zeta functions (Errata)*, 2022, arXiv:2203.04917.
[17] F. Faure, J. Sjöstrand, *Upper bound on the density of Ruelle resonances for Anosov flows*, Comm. Math. Phys. 308 (2011), no. 2, 325–364, doi:10.1007/s00220-011-1349-z, Primary preprint.
[18] S. Dyatlov, M. Zworski, *Dynamical zeta functions for Anosov flows via microlocal analysis*, Ann. Sci. Éc. Norm. Supér. (4) 49 (2016), no. 3, 543–577, Publisher text, doi:10.24033/asens.2290.
[19] Y. Guedes Bonthonneau, *Perturbation of Ruelle resonances and Faure–Sjöstrand anisotropic space*, Rev. Un. Mat. Argentina 61 (2020), no. 1, 63–72, doi:10.33044/revuma.v61n1a03, Primary preprint.
[20] N. V. Dang, G. Rivière, *Topology of Pollicott–Ruelle resonant states*, Ann. Sc. Norm. Super. Pisa Cl. Sci. (5) 21 (2020), special issue, 827–871, doi:10.2422/2036-2145.201804_010, Primary preprint, Theorem 2.1.
[21] S. Dyatlov, M. Zworski, *Ruelle zeta function at zero for surfaces*, Invent. Math. 210 (2017), no. 1, 211–229, Primary author PDF.
[22] M. Cekić, G. P. Paternain, *Resonant spaces for volume-preserving Anosov flows*, Pure Appl. Anal. 2 (2020), no. 4, 795–840, doi:10.2140/paa.2020.2.795, Primary preprint.
[23] M. Cekić and G. P. Paternain, *Resonant forms at zero for dissipative Anosov flows*, Geom. Topol. **29** (2025), no. 7, 3635–3716. doi:10.2140/gt.2025.29.3635; arXiv:2211.06255v2.
[24] N. V. Dang, C. Guillarmou, G. Rivière, and S. Shen, *The Fried conjecture in small dimensions*, Invent. Math. **220** (2020), no. 2, 525–579. doi:10.1007/s00222-019-00935-9; arXiv:1807.01189v3.
[25] M. Cekić, B. Delarue, S. Dyatlov, and G. P. Paternain, *The Ruelle zeta function at zero for nearly hyperbolic 3-manifolds*, Invent. Math. **229** (2022), no. 1, 303–394. doi:10.1007/s00222-022-01108-x; arXiv:2009.08558v3. Equation references follow arXiv version v3.
[26] Y. Chaubet and N. V. Dang, *Dynamical torsion for contact Anosov flows*, Anal. PDE **17** (2024), no. 8, 2619–2681. doi:10.2140/apde.2024.17.2619; arXiv:1911.09931v2. Author version v2 (equation numbering used here).
[27] J.-M. Bismut, S. Shen, *Anosov vector fields and Fried sections*, Comm. Math. Phys. 406 (2025), article 240, doi:10.1007/s00220-025-05400-8, Primary final author PDF.
[28] T. Humbert and Z. Tao, *Generic twisted Pollicott–Ruelle resonances and zeta function at zero*, preprint, arXiv:2602.12166v2, 3 March 2026 (first version posted 12 February 2026 under the title *Twisted Pollicott–Ruelle resonances and zeta function at zero on surfaces*).
[29] P. Spilioti, *Twisted Ruelle zeta function on hyperbolic manifolds and complex-valued analytic torsion*, J. Inst. Math. Jussieu 24 (2025), no. 3, 813–845, doi:10.1017/S1474748024000586, arXiv:2004.13474.
[30] S. Shen, *Complex valued analytic torsion and dynamical zeta function on locally symmetric spaces*, Int. Math. Res. Not. IMRN (2023), no. 5, 3676–3745, arXiv:2009.03427.
[31] M. Jézéquel and J. Zung, *Zeta functions and the Fried conjecture for smooth pseudo-Anosov flows*, preprint, arXiv:2409.17014v2, 18 August 2026 (first version posted 25 September 2024).
[32] S. Shen and J. Yu, *Morse–Smale flow, Milnor metric, and dynamical zeta function*, J. Éc. polytech. Math. 7 (2020), 585–607, doi:10.5802/jep.154.
[33] D. Margalit and S. Spallone, *A homological recipe for pseudo-Anosovs*, Primary author PDF, Propositions 1–2.
[34] W. P. Thurston, *Hyperbolic structures on 3-manifolds, II: surface groups and 3-manifolds which fiber over the circle*, Primary preprint.